

Cancer incidence and mortality in relation to body mass index in the Million Women Study: cohort study

Objective To examine the relation between body mass index (kg/m²) and cancer incidence and mortality.

Design Prospective cohort study. Participants 1.2 million UK women recruited into the Million Women Study, aged 50-64 during 1996-2001, and followed up, on average, for 5.4 years for cancer incidence and 7.0 years for cancer mortality. Main outcome measures Relative risks of incidence and mortality for all cancers, and for 17 specific types of cancer, according to body mass index, adjusted for age, geographical region, socioeconomic status, age at first birth, parity, smoking status, alcohol intake, physical activity, years since menopause, and use of hormone replacement therapy. Results 45 037 incident cancers and 17 203 deaths from cancer occurred over the follow-up period. Increasing body mass index was associated with an increased incidence of endometrial cancer (trend in relative risk per 10 units=2.89, 95% confidence interval 2.62 to 3.18), adenocarcinoma of the oesophagus (2.38, 1.59 to 3.56), kidney cancer (1.53, 1.27 to 1.84), leukaemia (1.50, 1.23 to 1.83), multiple myeloma (1.31, 1.04 to 1.65), pancreatic cancer (1.24, 1.03 to 1.48), non-Hodgkin's lymphoma (1.17, 1.03 to 1.34), ovarian cancer (1.14, 1.03 to 1.27), all cancers combined (1.12, 1.09 to 1.14), breast cancer in postmenopausal women (1.40, 1.31 to 1.49) and colorectal cancer in premenopausal women (1.61, 1.05 to 2.48). In general, the relation between body mass index and mortality was similar to that for incidence. For colorectal cancer, malignant melanoma, breast cancer, and endometrial cancer, the effect of body mass index on risk differed significantly according to menopausal status. Conclusions Increasing body mass index is associated with a significant increase in the risk of cancer for 10 out of 17 specific types examined. Among postmenopausal women in the UK, 5% of all cancers (about 6000 annually) are attributable to being overweight or obese. For endometrial cancer and adenocarcinoma of the oesophagus, body mass index represents a major modifiable risk factor; about half of all cases in postmenopausal women are attributable to overweight or obesity.